

GARGI RAMACHARAN

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Science in Electrical Engineering and Computer Science (EECS)

Expected Graduation: May 2029

- **Relevant Coursework:** Structure and Interpretation of Computer Programs, Data Structures, Signals and Systems, Great Ideas of Computer Architecture (Machine Structures), Discrete Mathematics and Probability Theory

EXPERIENCE

AP0110 (Berkeley SkyDeck)

San Francisco, CA

Software Engineering Intern

May 2026 – Aug 2026

- Built the patient and practitioner React dashboards for Edge-Med, a local-first health screening prototype, on a unified data layer that merges seeded records with locally stored check-ins and recomputes risk on every read.
- Integrated a locally-run Ollama LLM as the check-in chatbot's conversational and structured-extraction layer through a Vite dev proxy, keeping the deterministic rule engine as the sole risk scorer so no patient data leaves the machine.
- Built dependency-free SVG vitals charts (BP, HR, SpO2, temperature) with clinical threshold banding, plus an after-visit summary that traces each risk label to the metric and cutoff that triggered it.

Roku, Inc.

San Jose, CA

Software Engineering Intern

Summer 2024, Summer 2025

- Led 5 interns building a Roku SceneGraph quiz game (BrightScript): designed a scene-level state machine coordinating 8 custom XML components through observer-pattern field aliases, focus handoff, and remote-key routing.
- Offloaded HTTPS fetches to a background Task node with async message ports and timeout/response-code handling so quiz content loads at runtime from a JSON API.

PROJECTS

Cloudscape (Perplexity Hackathon First Place) | Next.js, TypeScript, AWS SDK v3

April 2026

- Built an AI-powered incident response tool that ingests CloudWatch alarms via SNS webhooks, then concurrently pulls metrics, logs, and CloudTrail events from AWS using SDK v3 with cross-account STS AssumeRole for read-only access.
- Integrated Perplexity Sonar Reasoning Pro to correlate AWS telemetry with real-time web search, returning structured root cause analysis, ranked remediation options with CLI/Terraform commands, and Google SRE-format post-mortems.

ShameWall | Chrome Extension (Manifest V3), JavaScript, Chart.js

Feb 2026

- Built a Chrome extension that runs a content script across all URLs to detect user-defined distracting sites and inject full-page HTML/CSS overlays into the DOM, blocking access without redirecting away from the page.
- Implemented webcam-based "shame" capture via the MediaDevices getUserMedia API rendered inside the block overlay; popup UI uses Chart.js + chrome.storage for local-only per-site analytics.

Multi-Class Risk Identifier Using Home Measurable Vitals | Python, Random Forests

Nov 2025

- Built two machine learning models (clinical and at-home) using Random Forests, achieving 95.0% and 92.5% accuracy respectively on 1,000 cleaned and encoded patient records.
- Engineered a home-testable model using only 4 vitals (HR, RR, temperature, O2 saturation), proving near-clinical accuracy with low-cost, widely accessible inputs; awarded Most Creative (Finalist) for feature engineering.

LEADERSHIP

Society of Women Engineers (SWE)

Berkeley, CA

HSEP Co-Chair | SWE++ Co-Chair

Aug 2025 – Present

- Co-designed and taught a 10-week Python curriculum for 25 local middle school students, covering basic syntax through complex loops; authored lesson plans and mini-lecture videos for weekly sessions.
- Managed a 25+ student classroom, providing 1-on-1 debugging support and translating high-level programming concepts into interactive activities.

SKILLS

Languages: Python, C, Java, C++, JavaScript, TypeScript, SQL

Frameworks: React, Next.js, Vite, Tailwind, Jest

Tools & Platforms: Git, Docker, Linux, PostgreSQL, AWS (CloudWatch, SNS, STS), Ollama, REST APIs